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WaveScopeDefTopic

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About WaveScope

About WaveScope

The WaveScope tool is an interactive display which plots waveform data. It allows you to visually inspect the resulting waveform on the WaveScope display without affecting the authenticity of the waveform’s data.

WaveScope allows you to:

- [Display sourced waveforms or captured DspWaves](#)
- [Display multiple waveforms on a scope simultaneously \(Multi-channel mode\)](#)
- [Superimpose multiple waveforms on a scope for comparison \(Overlay mode\)](#)
- [Display waveforms in either the time or frequency domain \(FFT and Inverse FFT\)](#)
- [Determine the exact data point value of a sample \(Text Display and Cursor mode\)](#)
- [Use crosshair cursors to display actual values at full precision \(Cursor mode\)](#)
- [Scale waveforms with user-friendly controls \(Zoom in, Zoom out\)](#)

- [Read in data from text files \(.txt\)](#), IMAGE wave files (.wav), or DspWaves (.plot).

The WaveScope information also includes the following sections:

- [Operating WaveScope](#)

- [WaveScope Toolbar and Command Reference](#)

WaveScope is often used with the WaveDesigner tool when creating waveforms. For more information, see [About WaveDesigner](#).



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WaveScope Toolbar and Command Reference

WaveScope Toolbar and Command Reference

The following sections describe the WaveScope menu commands and the actions they perform:

- | | |
|---|---|
| • | File Menu Commands |
| • | Edit Menu Commands |
| • | View Menu Commands |
| • | WaveScope Menu Commands |

	
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WaveScope Toolbar and Command Reference : File Menu Commands

File Menu Commands

File menu commands perform basic actions for operating the WaveScope:

Open	 (Ctrl+O) Opens a text (.txt), media wave (.wav), or a DspWave (.plot) file. See Opening a File.
Close WaveScope	 Closes the active WaveScope display. This command is not available when running WaveScope in standalone mode. See Closing a File and Exiting WaveScope.
Save	 Launches a dialog box, which lets you specify the waveform you want to save. You can save as a text file (.txt) or as an Image wav file (.wav). See Saving a File.
Close All	Closes WaveScope and all other tools open in TDE. See Closing a File and Exiting WaveScope.
Print	 Launches the standard Windows print dialog box, which only prints the WaveScope display area (including the axes).
Exit TDE	Closes all active TDE tools. This command is not available when running WaveScope in standalone mode. See Closing a File and Exiting WaveScope.

For more information, see [Operating WaveScope](#).

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WaveScope Toolbar and Command Reference : Edit Menu Commands

Edit Menu Commands

Edit menu commands choose a waveform to be modified or removed.

Undo Zoom	 (Ctrl+Z) Performs incremental unzoom. It will undo, in reverse order, any zoom ins or outs that were done. Grayed out if no zoom has been performed. See Undo Zoom.
Select Waveform	Selects a waveform. This command displays a submenu that lists all opened (displayed and hidden) waveforms. Select a waveform by clicking the waveform’s name in the submenu. The selected waveform will now have a check mark next to it. To deselect the waveform, click the waveform name and the check mark disappears. See Selecting a Waveform.
Copy to Clipboard	 (Ctrl+C) Copies the WaveScope to the clipboard.
Delete	 (Shift+Delete) Removes the selected waveform from WaveScope. See Hiding and Deleting a Waveform.
Delete All	 (Ctrl+Shift+Delete) Removes all waveforms from WaveScope. See Hiding and Deleting a Waveform.

For more information, see [Viewing Waveforms](#).

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WaveScope Toolbar and Command Reference : View Menu Commands

View Menu Commands

[View](#) menu commands allow you to change the perspective of the waveform that is plotted on the display. Some commands change the environmental settings of the WaveScope display (Zoom In, Set Axes), others change the visual style of the waveform (Cycle Through Formats), and others provide access to different modes (Multi-Channel, Overlay, Text Display).

Text Display	 (Ctrl+T) Displays waveform data values in a tabular form. Data values can be displayed in three formats: integer, hex, or binary. See Text Display Mode .
Hide Waveform	Selects a waveform to hide from the display. This command displays a submenu that lists all opened (displayed and hidden) waveforms. Select a waveform to hide by clicking a waveform’s name in the submenu. Each hidden waveform will now have a check mark next to it. To redisplay a waveform, click the waveform name and the check mark disappears. See Hiding and Deleting a Waveform .
ShowAllWaveforms	 (Ctrl+W) Displays all open waveforms (including hidden waveforms) and zooms out so that all points of all waveforms are displayed in the WaveScope display area. See Showing All Waveforms .
Cycle Through Formats	 (Ctrl+Y) Rotates through the waveform format styles (line, stepped, points, drop-down lines). See Changing Trace Style and Color .

Zoom In



(Ctrl+N)

Zooms in by approximately 10% for both axes and snaps to a logical value. The Page Down key also does this and zooms in by 50% when combined with the Ctrl key. See [Zooming In](#).

Zoom Out



(Ctrl+M)

Performs unzooms at 10% increments. The Page Up key on the keyboard also does this and zoom outs by 50% when combined with the Ctrl key. See [Zooming Out](#).

Zoom Settings

Sets the scale factor for zooms. See [Adjusting Zoom Settings](#).

Multi-Channel



Allows the WaveScope to display more than one wave trace at a time. The default is to display a single trace. See [Activating Multi-Channel Mode](#).

Overlay



Plots waveforms on top of each other in the display (the default). If Multi-channel mode is not active, Overlay is grayed out. See [Deactivating Overlay Mode](#).

Trace

File Information

Displays the header information for .wav files. Normally contains the file type, domain, sample size, and sample rate.

WaveformFFT



(Ctrl+F)

Performs an FFT on the selected waveform (time domain only). See [Performing an FFT](#).

WaveformInverseFFT



(Ctrl+I)

Performs an Inverse FFT on all active waveforms that are in the frequency domain. See [Performing an Inverse FFT](#).

of Cycles to Display

Specifies the number of repeated waveform cycles displayed in the scope. This offers a precise alternative to zooming in on a particular section of the waveform.

Format>Trace>Style	Specifies how the line style of the trace: • Line: Displays the selected waveform as a constant line by connecting all data points. • Stepped: Displays the selected waveform by connecting the points with stepped lines. • Points: Displays only data points of the waveform. • Dropdown Lines: Displays a line from each data point to the bottom of the WaveScope display. This is useful for viewing data in the frequency domain. See Changing Trace Style and Color.
Format>Trace>Color	Sets the color of the active trace. This launches the standard Microsoft color selector. See Changing Trace Style and Color.
Format>Set Axes	 (Ctrl+A) Launches a dialog box that allows the axes formats and the minimum and maximum X and Y axes values to be set. By default, these are set to autoscale. See Changing Axes Values.
Format>Grid Lines	(Ctrl+G) Launches a dialog box used to set the display grid line values and frequency. See Changing Grid Lines.

For more information, see:

- [Viewing Waveforms](#)
- [Changing WaveScope Display Properties](#)

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WaveScope Toolbar and Command Reference : WaveScope Menu Commands

WaveScope Menu Commands

WaveScope menu commands control some WaveScope display environmental modes. From the WaveScope menu, you can set the scope display to plot waveforms in the time domain or the frequency domain, and you can activate the WaveScope’s Cursors mode.

CompleteFFT	 (Ctrl+F) Performs an FFT on waveforms that are in the time domain. See Performing an FFT.
CompleteInverseFFT	 (Ctrl+I) Performs an inverse FFT on waveforms that are in the frequency domain. See Performing an Inverse FFT.
Cursors	 (Ctrl+K) Enables or disables Cursor mode which allows you to use hovering reference lines to perform calculations using precise data point values. See Using Cursors Mode.
Snap Cursors To Point	 When the cursors are active, this controls if cursor values are snapped the closest point on the waveform, or if they are free to go anywhere on the plot. This is the default when using Cursors Mode. See Snap Cursors to Point.
Options	
SaveWaveScopeSetup	Saves the current display preference settings so that they will be the default the next time WaveScope is opened. The preferences affected

are: trace color, axes values, grid lines, cursors, cursors snapped to point, and multi-channel and overlay modes.

Notes:

- This action is only available if WaveScope was accessed through a TDE tool.
- The trace color is saved by default in regular use. Therefore, you need not use this command if you only want to save the trace color.

For more information, see:

- | | |
|---|---|
| • | Viewing Waveforms |
| • | Changing WaveScope Display Properties |





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Operating WaveScope

Operating WaveScope

This section describes the basic procedures used to operate the WaveScope tools. Most procedures are intuitive based on standard Windows practices, but these sections also describe advanced and alternative ways of accomplishing these common tasks.

This section includes the following topics:

- | | |
|---|---|
| • | Starting WaveScope |
| • | Opening a File |
| • | Saving a File |
| • | Closing a File and Exiting WaveScope |
| • | Viewing Waveforms |
| • | Changing WaveScope Display Properties |

	
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Operating WaveScope : Starting WaveScope

Starting WaveScope

Start the WaveScope tool through [TDE Tools](#), [Mixed Signal Workshop](#), [WaveDesigner](#), [DspWave](#), or [as a stand-alone executable file](#) (WaveScope.exe). Although the path to WaveScope may be different, the appearance and functionality is the same.

The following sections describe how to start WaveScope from different points of origin:

- | | |
|---|--|
| • | Starting WaveScope Through TDE Tools |
| • | Starting WaveScope Through WaveDesigner |
| • | Starting WaveScope Through Mixed Signal Workshop |
| • | Starting WaveScope Through DspWave |
| • | Starting WaveScope Through a WaveScope Executable File |



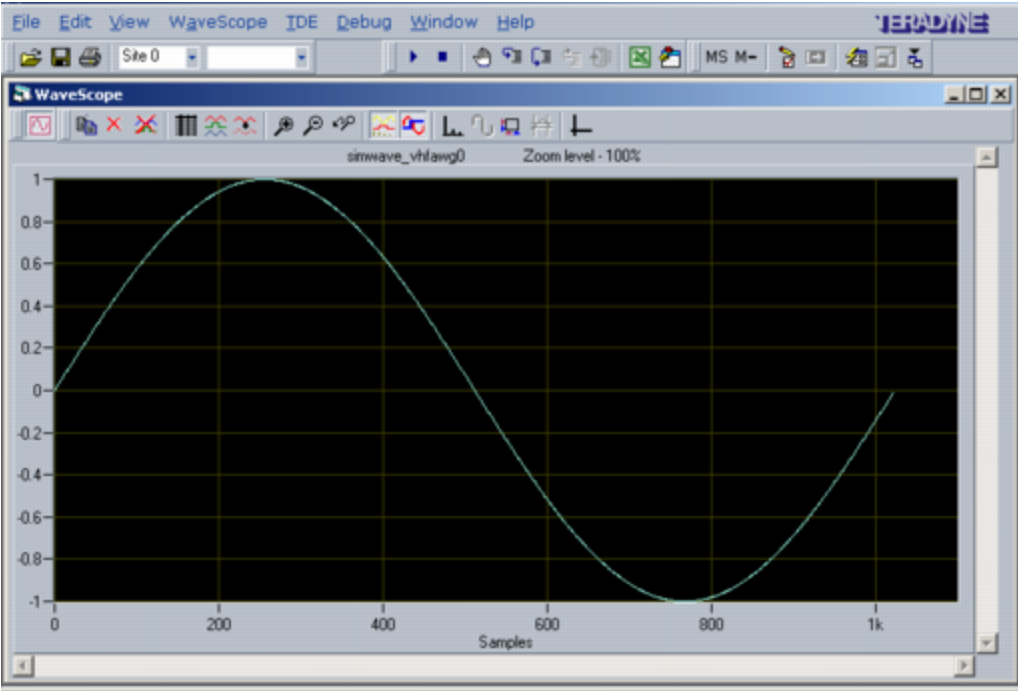
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Operating WaveScope : Starting WaveScope : Starting WaveScope Through TDE Tools

Starting WaveScope Through TDE Tools

Start WaveScope from the Test Debug Environment (TDE) by selecting TDE>Tools>WaveScope. You can also start WaveScope by selecting Tools>WaveScope in the Launch group on the IG-XL Tools tab. This starts TDE (if it is not already open) and opens WaveScope.



WaveScope Tool Display in TDE

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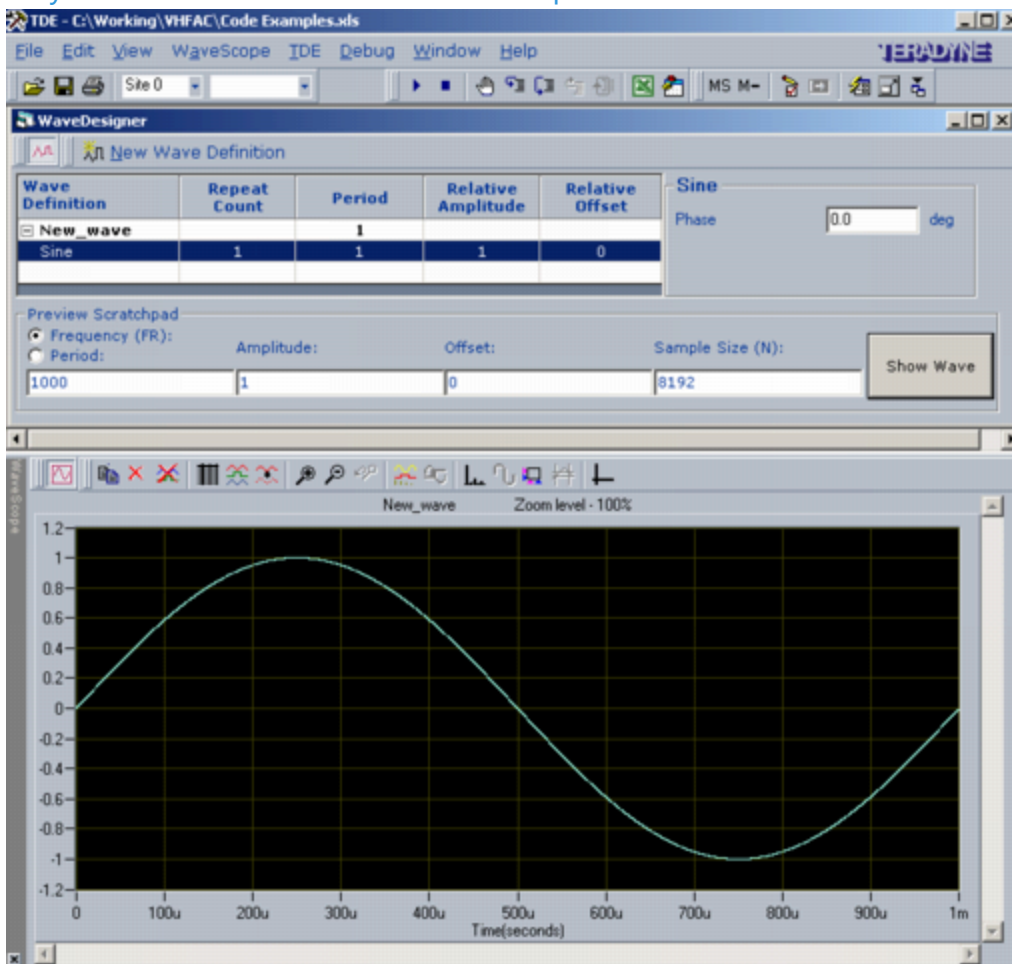
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Operating WaveScope : Starting WaveScope : Starting WaveScope Through WaveDesigner

Starting WaveScope Through WaveDesigner

The bottom half of the WaveDesigner tool display contains the WaveScope tool. While its main purpose in this environment is to display waveforms that you create using the WaveDesigner, it is a fully functional version of the WaveScope that can read in files external to the WaveDesigner.



WaveScope Integrated into the Lower Portion of WaveDesigner

See [WaveDesigner Display](#) for instructions on starting the WaveDesigner tool.

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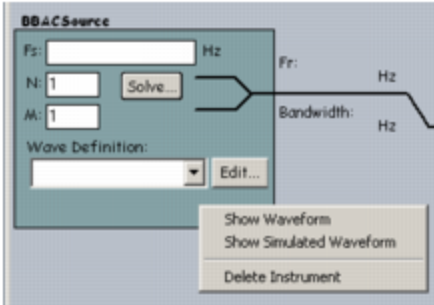
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Operating WaveScope : Starting WaveScope : Starting WaveScope Through Mixed Signal Workshop

Starting WaveScope Through Mixed Signal Workshop

You can start WaveScope from within the Mixed Signal Workshop. With a wave definition specified in the active Source instrument block, right-click the source instrument block and select either Show Waveform or Show Simulated Waveform.

Show Waveform	Displays the shape of the waveform built directly from the wave definition samples.
Show Simulated Waveform	Displays a simulated waveform that is built from the raw wave definition samples that have been passed through and treated by the instrument.



Previewing a Sourced Waveform from MSW Using WaveScope

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Operating WaveScope : Starting WaveScope : Starting WaveScope Through DspWave

Starting WaveScope Through DspWave

WaveScope can display the resulting waveform of a DspWave procedure. The standalone WaveScope automatically starts when the procedure is run if you specify a .plot file (dspwavename.plot) in the VBT module.

The following syntax example creates a DspWave, fills it with data, and calls .plot, which brings up the WaveScope and displays the resulting waveform:

Private Sub TestPlot()

Dim MyWave As New DspWave

Dim Samples(1024) As Double

Dim i As Integer

For i = 0 To 1023

Samples(i) = Sin(2 * 3.1415926535898 / 1024 * i * (1 + 0.000000001))

Next

MyWave.Data = Samples

MyWave.Name = "Smeared Sine"

MyWave.Domain = TimeDomain

MyWave.Plot

End Sub



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Operating WaveScope : Starting WaveScope : Starting WaveScope Through a WaveScope Executable File

Starting WaveScope Through a WaveScope Executable File

You can start WaveScope by running an executable file located in the IG-XL \bin directory. The WaveScope.exe file is a part of the standard IG-XL installation, and it allows you to start WaveScope without having IG-XL open. Access this file from the Run command prompt in your computer's Start menu or by double-clicking the file name in the IG-XL \bin directory.

You can also start WaveScope by typing wavescope in the Command Prompt window.

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Operating WaveScope : Opening a File

Opening a File

WaveScope can read in data from files saved in text (.txt) or wave (.wav) formats. The waveform data contained in the file is read into the WaveScope and graphically plotted in the WaveScope display area.

Two common ways of reading in a file are using the File>Open command and clicking the Open button on the tool bar. When starting WaveScope from a DspWave procedure or from the Mixed Signal Workshop, the source data is automatically read in from the application.

WaveScope's default mode is to open and display one file at a time. To open and display more than one waveform, use multi-channel mode. See [Using Multi-Channel and Overlay Mode](#).

Using the Open Command in the File Menu

To open a file containing a waveform trace:

1. [Select File>Open](#).
The Open dialog box appears.
2. [Navigate to the folder that contains the desired file and select the file's name](#).
3. [Click the](#) Open button.
The waveform automatically appears in the WaveScope display area.

Using the Open Tool Bar Button

To open a file containing a waveform trace:

1. [Click the](#) Open button  on the WaveScope toolbar.
The Open dialog box appears

2.

Navigate to the folder that contains the desired file and select the file’s name.
3.

Click the Open button.

The waveform automatically appears in the WaveScope display area.

Automatically Read in Data

In most case, you use the Open button or command to read in a file. In some situations, however, WaveScope automatically reads in the waveform data from a adjoining source. For example, when a .plot is specified in the VBT module and the DspWave procedure is run, the WaveScope automatically receives the data and plots the resulting waveform. Similarly, Mixed Signal Workshop’s Show Waveform or Show Simulated Waveform commands automatically read in the source data from the specified wave definition and plot it on the WaveScope display.

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Operating WaveScope : Saving a File

Saving a File

In a typical WaveScope session, you use the WaveScope commands to manipulate the perspective of the plotted waveform to inspect the quality and characteristics of the waveform. You can save the plotted waveform for future reference.

Although a WaveScope session can display multiple waveforms simultaneously using multi-channel mode, each waveform is saved individually.

Using the Save Command in the File Menu

To save a waveform trace:

- | | |
|----|--|
| 1. | Select File>Save.
The Waveform Name dialog box opens. |
| 2. | Select the waveform trace to be saved and click the Apply button.
The Save As dialog box opens. |
| 3. | Specify the file name and the location of where the file is to be saved. |
| 4. | Click the Save button.
You can save a waveform file either as a text file (.txt) type or as an Image .wav file (.wav) type. |

Using the Save Button on the Tool Bar

To save a waveform trace:

1. Click the Save button  on the WaveScope toolbar.
The Waveform Name dialog box opens.
2. Select the waveform trace to be saved and click the Apply button.
The Save As dialog box opens.
3. Specify the file name and the location of where the file is to be saved.
4. Click the Save button.
You can save a waveform file either as a text file (.txt) type or as an Image .wav file (.wav) type.

	
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Operating WaveScope : Closing a File and Exiting WaveScope

Closing a File and Exiting WaveScope

When you close a file in WaveScope, you also close the WaveScope tool. Commands are also available that allow you to quit not only WaveScope, but all other tools and TDE in one action.

- [Using the Close Command in the File Menu](#)
- [Using the Close All Menu Command in the File Menu](#)
- [Using the Exit TDE Menu Command in the File Menu](#)
- [Using the Close Button in the WaveScope and TDE Window](#)

To close an active file and open another file immediately in WaveScope, use the [Open](#) command to replace the active file with a new file. This technique closes the active file without quitting WaveScope. See [Using the Open Command in the File Menu](#) for instructions on using the Open command.

Using the Close Command in the File Menu

The [File>Close](#) command closes the active WaveScope window, but it does not close the TDE window or any other TDE tools that are open.

To close the file, select [File>Close Wavescope](#). The WaveScope window closes immediately, but the TDE window remains open.

Using the Close All Menu Command in the File Menu

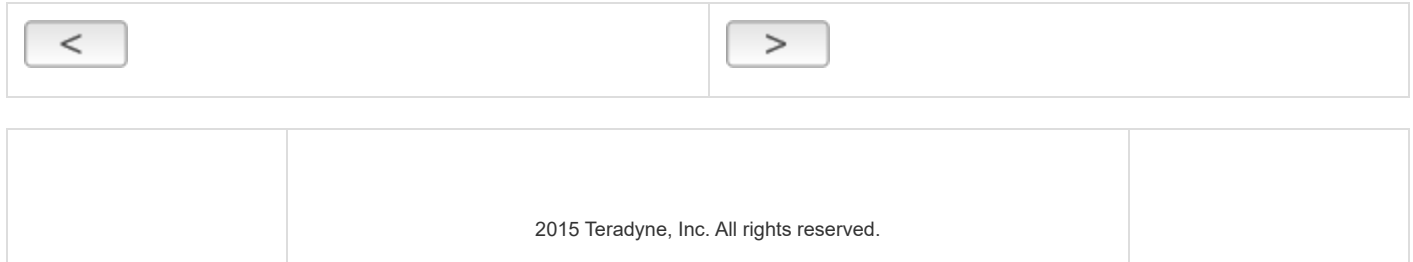
The [File>Close All](#) command closes all of the instances of WaveScope opened using the [DspWave.Plot](#) command. This command does nothing when WaveScope is started from within TDE or as a standalone WaveScope.

Using the Exit TDE Menu Command in the File Menu

The File>Exit TDE command closes the active WaveScope window, all other TDE tools that are open, and the active TDE window, all in one action.

Using the Close Button in the WaveScope and TDE Window

Click the standard Windows Close button  to close the WaveScope window or the TDE window.



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Operating WaveScope : Viewing Waveforms

Viewing Waveforms

You use the WaveScope tool to view waveforms that were created from waveform data. The following sections describe the ways in which you can control the scope's display for viewing waveform and sample data information:

- [Using Multi-Channel and Overlay Mode](#)

- [Selecting a Waveform](#)

- [Hiding and Deleting a Waveform](#)

- [Showing All Waveforms](#)

- [Scaling the View of a Waveform Trace](#)

- [Performing an FFT](#)

- [Performing an Inverse FFT](#)

- [Using Cursors Mode](#)

- [Text Display Mode](#)

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Operating WaveScope : Viewing Waveforms : Using Multi-Channel and Overlay Mode

Using Multi-Channel and Overlay Mode

Multi-channel mode allows you to view multiple waveform traces simultaneously on the WaveScope display. In its default state, multi-channel mode uses Overlay mode and superimposes the opened waveforms one on top of another. By deactivating Overlay mode, WaveScope separates the waveforms so that they appear one above the other (vertically) on the same display.

Activating Multi-Channel Mode

Multi-channel mode is deactivated in WaveScope default state. This means that unless you activate multi-channel mode, you can only display one waveform at a time.

To activate Multi-channel mode, select View>Multi-Channel. Note that traces overlap each other in multi-channel mode by default. Use multi-channel mode and overlay mode to separate multiple waveforms on the display.

To deactivate Multi-channel mode, select View>Multi-Channel. Note that if you have more than one trace displayed, deactivating multi-channel mode deletes all traces except the currently selected trace. (You get a warning if you do this.)

Deactivating Overlay Mode

When you activate Overlay mode (the default when you activate Multi-Channel mode), waveforms appear superimposed on the display. Note that you can only use Overlay mode after you activate Multi-channel mode.

You can deactivate Overlay mode to separate multiple traces within the same WaveScope display (so that each is displayed separately from the others).

To deactivate Overlay mode:

- With two or more traces plotted in the scope display, select View>Overlay. The traces now appear one above the other.

Note that multiple sets of y axis values appear in the margin each corresponding to one trace.

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Operating WaveScope : Viewing Waveforms : Selecting a Waveform

Selecting a Waveform

When only one trace is open in the WaveScope display, all commands apply to that trace. When multiple traces are open, however, you must select the trace that you want to use.

The following topics describe how to select a waveform trace on the scope display:

- [Using the Select Waveform Command in the Edit Menu](#)

- [Using Keyboard Shortcuts to Select a Waveform](#)

Note that clicking a trace in the WaveScope display is not a valid method for selecting a trace. Also note that you cannot deselect a trace (make the trace thinner) once you select it. You must select another open waveform trace.

Using the Select Waveform Command in the Edit Menu

To select a waveform on the scope display:

1. [Select](#) Edit>Select Waveform.

This command displays a submenu that lists all opened (displayed and hidden) waveforms.

2. [Choose](#) the name of the trace from the list of waveforms in the submenu.

The waveform trace selected now appears as a thicker line in the display.

The selected waveform will now have a check mark next to it in the menu. To deselect the waveform, click the waveform name and the check mark disappears.

Note that although hidden waveforms are listed in the submenu, selecting a hidden waveform does not make it reappear in the scope display. See [Show All Waveforms in the View Menu Commands](#)

section.

Using Keyboard Shortcuts to Select a Waveform

To select a waveform trace, press the Tab key which cycles through each waveform that is open in the scope display.

		
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Operating WaveScope : Viewing Waveforms : Hiding and Deleting a Waveform

Hiding and Deleting a Waveform

If you don't want a trace to appear in the WaveScope display, you can either hide that trace or delete the trace from the display. Hiding a trace does not close the waveform file, it merely hides the trace so that it does not appear on the display.

To hide a waveform:

1. [Select View>Hide Waveform.](#)
2. [Choose the waveform from the pull-right menu.](#)

The hidden waveform will now have a check next to it in the pull-right menu. Note that the cleared trace has not been deleted, it is simply hidden.

To show a hidden waveform in the display again, do either of the following:

- [Select View>Hide Waveform](#) and select the hidden waveform to display.
- [Use the Show All Waveforms command](#) as described in [Showing All Waveforms](#).

To delete a single waveform from the display:

1. [Select Edit>Delete.](#)
You get a warning that delete is not reversible.
2. [Click Yes.](#)
The trace disappears from the WaveScope display.

To delete a waveform when two or more are displayed

1.

Select Edit>Delete.

A dialog box comes up that allows you to select the waveform to delete.
2.

Select the waveform to be deleted and click Yes.

You get a warning that delete is not reversible.
3.

Click Yes.

The selected trace is deleted from the WaveScope display.

To delete all waveforms from the display area, use the Edit>Delete All command.

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Operating WaveScope : Viewing Waveforms : Showing All Waveforms

Showing All Waveforms

The ShowAllWaveforms command displays all open waveforms (even those that have been cleared) and zooms out so that all points of all waveforms are displayed in the WaveScope display area. To show all waveforms, select View>ShowAllWaveforms, or click the ShowAllWaveforms button  in the tool bar.

All open traces appear in the display. Display settings, like zoom and axes values, are reset to encompass the largest open trace.

		
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Operating WaveScope : Viewing Waveforms : Scaling the View of a Waveform Trace

Scaling the View of a Waveform Trace

You can control the visual perspective of a waveform by using the zoom commands to scale the waveform display. You can then examine the waveform's shape characteristics in fine or expanded view.

WaveScope includes the following scaling functions:

- [Zooming In](#)
- [Zooming Out](#)
- [Undo Zoom](#)
- [Adjusting Zoom Settings](#)

Another way to control the view of a waveform is to change the values of the x and y axes. See [Changing Axes Values](#).

Zooming In

When a waveform trace is opened, the WaveScope display area (including the axes) automatically adjusts so that the entire waveform trace is visible in the scope area. To view the characteristics of a trace in greater detail, you can magnify, or zoom in on, certain sections of the trace.

The ways to zoom in on a trace are:

- [Click and drag around an area of the display that you want to magnify.](#)

- [Use the View>Zoom In menu command.](#)
- [Hold the Ctrl key down while using the Zoom In menu command to zoom in by 50%. See \[Using the Page Down Key to Zoom In\]\(#\).](#)
- [Use the Zoom In tool bar button](#)
- [Press the Page Down key. See \[Using the Page Down Key to Zoom In\]\(#\).](#)
- [Hold the Ctrl key down while pressing the Page Down key to Zoom In by 50%. See \[Using the Page Down Key to Zoom In\]\(#\).](#)

[Using the Page Down Key to Zoom In](#)

The WaveScope display automatically adjusts the trace, grid lines, and axes to a magnified view by 10% of the previous view.

Each additional Zoom In magnifies the view by 10% of the previous view. To Zoom In by 50% in one action, hold the Ctrl key while pressing the Page Down key.

[Zooming Out](#)

When a waveform trace is opened, the WaveScope display area (including the axes) automatically adjusts so that the entire waveform trace is visible in the scope area. Use the Zoom Out command to view areas beyond a trace's boundaries. You can also use the Zoom Out command to expand the display view after performing a Zoom In. See [Using the Page Up Key to Zoom Out](#) for instructions on how to widen and distance the scope's view perspective.

[The ways to Zoom Out on a trace are:](#)

- [Use the View>Zoom Out menu command to Zoom In](#)
- [Use the Zoom Out tool bar button](#)
- [Press the Page Up key. See \[Using the Page Up Key to Zoom Out\]\(#\).](#)
- [Hold the Ctrl key down while pressing the Page Up key to Zoom Out by 50%. See \[Using the Page Up Key to Zoom Out\]\(#\).](#)

[Using the Page Up Key to Zoom Out](#)

The WaveScope display automatically adjusts the trace, grid lines, and axes to expand by 10% of the previous view.

Each additional Zoom Out expands the view by 10% of the previous view. To zoom out by 50% in one action, hold the Ctrl key while pressing the Page Up key.

Undo Zoom

To undo the last zoom command using the Undo Zoom command, do any of the following:

• Select Edit>Undo Zoom

• Press Ctrl+Z

• Click the Undo Zoom button 

The WaveScope display automatically adjusts the trace, grid lines, and axes to the previous view.

Adjusting Zoom Settings

Use the Zoom Settings menu item to set the scalefactor for Zoom In and Zoom Out. For example, setting the scalefactor to 20 will zoom in the waveform from 100% to 120% in one click.

The default scalefactor is 10.

 **Note:**

The scalefactor is not the zoom percentage.





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Operating WaveScope : Viewing Waveforms : Performing an FFT

Performing an FFT

By default, WaveScope plots waveforms in the time domain so that the x axis represents the number of samples over time and the y axis represents voltage.

Waveforms displayed in the time domain can be regenerated in the frequency domain by performing a Fast Fourier Transform (FFT) operation. An FFT is a process by which digital information is converted from the time domain to the frequency domain for rapid spectral analysis. WaveScope automates the FFT process and plots the waveform in the frequency domain where the x axis measures the frequency and the y axis is either the linear or logarithmic measurement of the signal's power.

To perform an FFT operation on a waveform displayed in the time domain, do any of the following:

- [Select WaveScope>CompleteFFT.](#)
- [Press Ctrl+F.](#)
- [Click the FFT toolbar button !\[\]\(e27c4336460e9e6729a19580c0456728_img.jpg\).](#)
- [Press the right mouse button while the pointer is over the display area and select CompleteFFT.](#)

If more than one waveform is displayed, this command performs the FFT on all waveforms displayed. To perform an FFT on only a selected trace, leaving the remaining traces in the frequency domain:

1. [Select the trace to be transformed using the](#) Edit>Select Waveform command:

a. [Select the](#) Edit>Select Waveform command.

b. [Select the waveform.](#)
[A check mark appears next to the waveform, and the waveform trace specified appears thicker in the scope display.](#)

2. [Select](#) View>Trace>WaveformFFT.

			
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Operating WaveScope : Viewing Waveforms : Performing an Inverse FFT

Performing an Inverse FFT

Waveforms displayed in the frequency domain can be regenerated in the time domain by performing an inverse Fast Fourier Transform (Inverse FFT) operation.

To perform an inverse FFT operation on a waveform displayed in the scope's frequency domain, do any of the following:

- [Select](#) WaveScope>CompleteInverseFFT.
- [Press](#) Ctrl+I.
- [Click the](#) Inverse FFT toolbar button .
- [Press the right mouse button](#) while the pointer is over the display area and select CompleteInverseFFT.

If more than one waveform is displayed, this command performs an inverse FFT on all waveforms displayed. To perform an Inverse FFT on only a selected trace, leaving the remaining traces in the frequency domain:

1. [Select the trace to be transformed using the](#) Edit>Select Waveform command:

- a. [Select the](#) Edit>Select Waveform command.

- b.

Select the waveform.

A check mark appears next to the waveform, and the waveform trace specified appears thicker in the scope display.

2.

Select View>Trace>WaveformInverseFFT.

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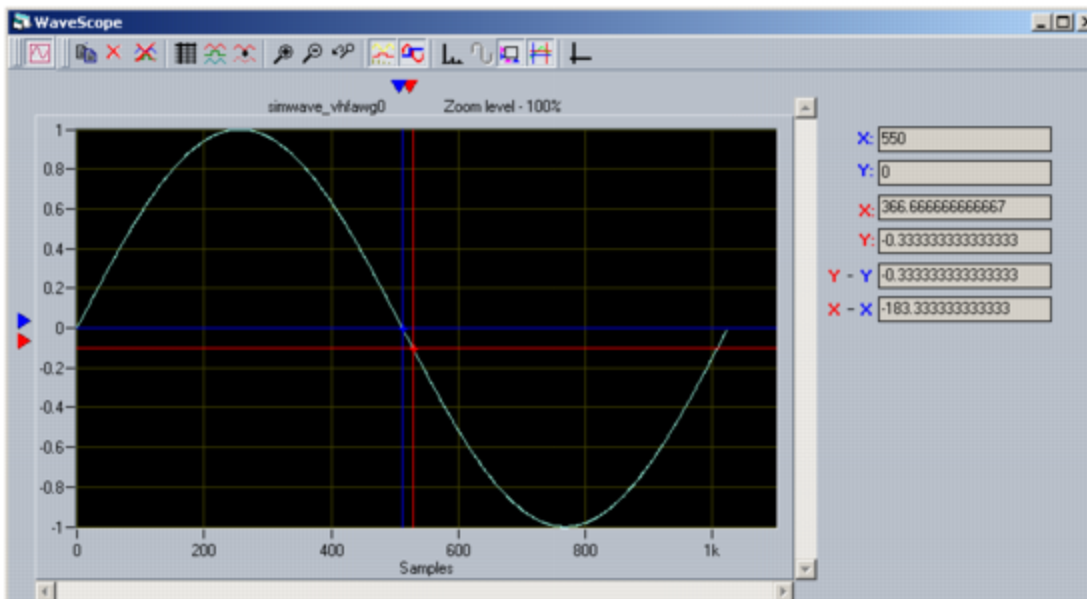
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Operating WaveScope : Viewing Waveforms : Using Cursors Mode

Using Cursors Mode

Cursors mode allows you to find the exact sample data along the waveform trace by sliding crosshair cursors over the waveform display. The figure shows the red and blue sliding cursors along the x axis and y axis of the WaveScope display, and the series of boxes on the right part of the window which track the cursors' intersection value along the waveform.



WaveScope Cursors Mode

To use Cursors Mode:

1. With a waveform trace open in the WaveScope display, do one of the following:

- Select WaveScope>Cursors.

- Click the Cursors button .

- Press Ctrl+K.

The WaveScope display changes to include a display area on the left and several x and y axes element boxes on the right.

Along the left and top margins of the display are two arrows (red and blue triangles) with associated lines that run through the display. These arrows and lines are cursors and allow you to determine the exact data value at the intersection of the cursor line and the waveform trace.

2. Select WaveScope>Snap Cursors to Point.

This turns off Snap, which is the default when using Cursor Mode.

3. Drag each arrow along the display's margin.

Note that as you drag the cursors across the display margins, the values change in the x and y axes element boxes at the right. These values represent the exact data value at the point of intersection of the trace and the cursor.

4. Drag only the blue x-axis cursor.

Note that not only does the blue x axis element value change on the right, but so does the element labeled X-X. The value in this element represents the difference between the data point under the red x-axis cursor and the data point under the blue x-axis cursor.

Snap Cursors to Point

Rather than trying to move the cursors on top of a data point manually, you can use the Snap Cursors to Point command. (This is the default when using Cursors Mode.) This command moves the cursors directly on top of the closest data point giving you the most precise data element values.

To snap cursors to point:

1. Select WaveScope>Snap Cursors to Point.

The cursors automatically move and intersect on the closest data point, producing the exact value of that data point.

2. While the Snap Cursors to Point is active, drag the cursors along the display margins.

Note that the cursors do not move freely as before. The cursors now move together and jump from one data point to another.

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Operating WaveScope : Viewing Waveforms : Text Display Mode

Text Display Mode

Text Display is a tool in WaveScope which allows you to view waveform data displayed as values. To view the values:

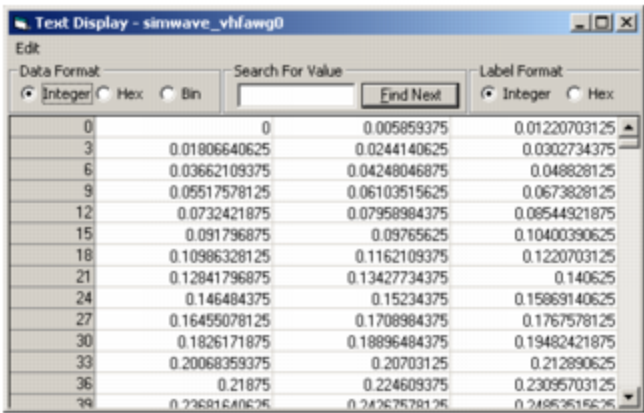
1.

With a trace open in the WaveScope display, select View>Text Display.

The Text Display dialog box appears and contains a table of waveform data values.
2.

To resize the dialog, drag the outside frame of the dialog box or click the Maximize button  in the upper right corner of the dialog box.

Note that the table columns automatically wrap in relation of the size of the dialog box.



Text Display

This tool includes the following controls:

Data Format

Selects format to view values: Integer (default, displays doubles), Hex, or Binary.

Search For Value	Allows you to search for a particular value.
Find Next	Finds the next value specified in the Search for Value box.
Label Format	Selects the label format in the left column (offset from zero) as Integer (default) or Hex.

Following are options in the Edit menu:

Select All	(Ctrl + A) Selects all values displayed in the grid. Also available in the right-click menu in the table.
Copy	(Ctrl + C) Copies grid values to the clipboard. Also available in the right-click menu in the table.

Searching for a Value

To search for a sample data value:

1.

Start to type the value into the Search For Value box.

The tool automatically starts searching among the data cells for the value based on your input.
2.

Once a value is found, press Enter on the keyboard to search for the next instance of that value.

The left hand column shows the offset from zero of the first value for that row. This label margin can be displayed in integer or hex format.

You cannot edit the values in the text display cells since there is no interaction between the Text Display and the WaveScope.

	
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Operating WaveScope : Changing WaveScope Display Properties

Changing WaveScope Display Properties

The WaveScope display includes some environmental features to help reference and scale waveform traces, including axes values, grid lines, and the color and style of traces. You can change the properties of these display features to get a better perspective on a waveform’s characteristics. The following topics describe how to change WaveScope display properties:

- | | |
|---|---|
| • | Changing Axes Values |
| • | Changing Grid Lines |
| • | Changing Trace Style and Color |
| • | Saving WaveScope Display Properties |

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Operating WaveScope : Changing WaveScope Display Properties : Changing Axes Values

Changing Axes Values

WaveScope plots waveforms in relation to a standard x and y axis graph. The default axes settings automatically scale the axes to the size of the largest trace open in the display and display linear units in the time domain.

You can change the axes settings to span a particular range or to be measured in a particular unit by editing the following parameters in the Axes Values dialog box. See [Changing the X Axis Values](#) and [Changing the Y Axis Values](#) for instructions on changing the axes values.

Axes Units	Sets the units for the x and y axes: X Axis: Sets the unit of measurement for the x axis. You can set the units to Samples/Bin (default) or Seconds/Hertz. Y Axis: Sets the unit of measurement for the y axis. You can set the units to Linear (default), 20log20, or 10Log10.
Axes Scaling: Manual Scaling	Allows you to enter the exact minimum and maximum values for the axes settings. By default, this option is unchecked, and the entire waveform is visible in the display area. Min X: Sets the display's cutoff value on the left side of the x axis. Max X: Sets the display's cutoff value on the right side of the x axis.

- **Min Y:** Sets the display's cutoff value for the lowest value along the y axis.
- **Max Y:** Sets the display's cutoff value for the highest value along the y axis.

Changing the X Axis Values

The X axis can display frequency or time values.

To change the axes values:

1. **Select** View>Format>Set Axes or press the Set Axes button  on the toolbar.

The Axes Values dialog box appears which displays the default settings (Manual Scaling unchecked).

2. **Check the** Manual Scaling check box.

This allows you to enter minimum and maximum axes values for each active trace displayed.

Change the Max X value by clicking in the text box and typing in a value that is greater than the sample size of the waveform trace.

3. **Click the** Apply button to apply the changes to the axes in the active WaveScope display.

4. **Click** OK.

Changing the Y Axis Values

The display of the y-axis depends on whether WaveScope is in Overlay mode. See Using Multi-Channel and Overlay Mode for a description of Overlay mode.

If Overlay mode is not active, all waveforms share a single y axis. By default, it is zoomed out to just above and beyond the largest and smallest values on any of the waveforms, so that nothing is cut off in the display.

If Overlay mode is not active, each waveform has a different y axis.

To set the values for the y axis:

1. **Select** View>Format>Set Axes or press the Set Axes button  on the toolbar.

By default, both axes are not set to Manual Scaling, which disables the ability to set specific maximum and minimum y axis values.

2.

Check the Manual Scaling check box to enable the y axis value boxes.
3.

Click the Max Y and Min Y boxes. Enter the value from which the associated trace will be referenced to along the y axis.
4.

By default the y axis is displayed in linear units. Change the y axis to display in either 20Log10 or 10Log10 format by clicking the appropriate buttons in the Axes Units frame.
5.

To apply changes to the axes on the WaveScope, click the Apply button.
6.

Click OK.

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Operating WaveScope : Changing WaveScope Display Properties : Changing Grid Lines

Changing Grid Lines

WaveScope’s display area is referenced by background grid lines. By default the grid lines are lined up to the major tick marks along both axes. You can change the grid lines to be set on Major and/or Minor ticks marks or based on a particular increment value (for example, every tenth sample).

To change the grid lines frequency for each axis, edit the following parameters in the Grid Settings dialog box (View>Format>Grid Lines):

Major	When selected, grid lines appear at major, whole axis unit, intervals in the WaveScope display.
Major/Minor	Specifies whether waveform trace is referenced by grid lines on the whole unit axis value (major), the half axis unit (minor), or both.
Increment	Specifies the sample interval value at which a grid line will appear.
Increment, Multiplier	Indicates the number of samples between the grid lines on the WaveScope display and any multiplier. This control is active only if Increment is selected.

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Operating WaveScope : Changing WaveScope Display Properties : Changing Trace Style and Color

Changing Trace Style and Color

You can display waveform traces in one of four styles and in a variety of colors. The default style is Line, but you can change a trace’s style to Stepped, Points, or Dropdown Lines. See [View Menu Commands](#) for a description of the styles.

To change a trace style:

1. [Select the trace whose style is to be changed using](#) Edit>Select Waveform.

The waveform trace specified appears thicker in the scope display.

2. [Select](#) View>Format>Trace>Style.

3. [Select the desired style:](#) Line, Stepped, Points, or DropDown.

You can also use WaveScope’s View>Cycle through Formats command or click the Trace Format button . This command rotates the selected trace through the four styles.

To change the color of a trace:

1. [Select the trace whose color is to be changed using](#) Edit>Select Waveform.

The waveform trace specified appears thicker in the scope display.

2. [Select](#) View>Format>Trace color.

A color palette window appears

3. Select a basic color or create your own custom color using the palette spectrum and color values.

4. Click OK.

The trace displays in the color specified.

Note that the specified color of the primary waveform becomes the default color of future primary traces until you change the color again.



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Operating WaveScope : Changing WaveScope Display Properties : Saving WaveScope Display Properties

Saving WaveScope Display Properties

You can save certain WaveScope settings so that the next time you start the tool, the settings remain as you had left them. Property settings that you can change and then save include:

- | | |
|---|-----------------------|
| • | Trace color |
| • | Axes values |
| • | Multi-channel mode |
| • | Overlay mode |
| • | Cursors mode |
| • | Cursors snap to point |
| • | Grid line values |

To save these WaveScope settings, make changes to the settings and then select WaveScope>Options>SaveWaveScopeSetup.

Note that color changes to the waveform trace are saved by default, making this procedure unnecessary if that is the only change to be saved.

This option only works when WaveScope is started from within TDE. Saved changes to the WaveScope property settings are saved to the tool itself. These changes are not isolated to the active test program. They also apply to all future sessions where WaveScope is used, even when using a completely different test program or DataTool has been restarted.

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